DNA Methylation Analysis Report

Differential Methylation in Rheumatoid Arthritis

Dataset: GSE42861 | Platform: Illumina HumanMethylation450 BeadChip

Organism: Homo sapiens | Tissue: Peripheral Blood Leukocytes (PBLs)

Tools Used: ChAMP | minfi | SeSAMe

Samples: 5 RA Cases | 5 Normal Controls

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1. Introduction

Long-term autoimmune rheumatoid arthritis (RA) is typified by systemic involvement and inflammatory joint damage. By modifying gene expression in immune cells without changing the DNA sequence itself, epigenetic modifications—specifically, changes in DNA methylation—have been linked to the pathogenesis of RA.
This study uses information from the GEO dataset GSE42861 to examine genome-wide DNA methylation changes between RA patients and healthy controls. The work uses Illumina HumanMethylation450 (450K) BeadChip arrays to profile bisulphite-converted DNA from peripheral blood leukocytes (PBLs) and examine approximately 485,000 CpG sites throughout the human genome.
For analysis, three complimentary R-based pipelines were employed:
• ChAMP (Chip Analysis Methylation Pipeline): for DMP identification, normalization, and quality assurance

• minfi—for differential methylation analysis, independent quality control, and normalizing validation

- SeSAMe — for IDAT-level processing with improved signal correction

2. Dataset Description

The dataset used in this study is publicly available through the NCBI Gene Expression Omnibus (GEO) database.

Parameter	Details
GEO Series	GSE42861
Title	Differential DNA methylation in Rheumatoid Arthritis
Organism	Homo sapiens
Tissue	Peripheral Blood Leukocytes (PBLs)
Platform	Illumina HumanMethylation450 BeadChip (GPL13534)
Experiment Type	Methylation profiling by array
Release Date	January 21, 2013
Samples Used	10 (5 RA cases, 5 normal controls) — subset

2.1 Sample Sheet

The following 10 samples (a representative subset from the full cohort) were included in this analysis:

Sample (GSM)	Sentrix ID	Sentrix Position	Group	Array
GSM1051525	7800246024	R01C01	Disease (RA)	Array 1
GSM1051526	7800246024	R02C01	Disease (RA)	Array 1
GSM1051527	7800246024	R03C01	Disease (RA)	Array 1
GSM1051528	7800246024	R04C01	Disease (RA)	Array 1

Sample (GSM)	Sentrix ID	Sentrix Position	Group	Array
GSM1051529	7800246024	R05C01	Disease (RA)	Array 1
GSM1051533	7800246024	R03C02	Control	Array 1
GSM1051534	7800246024	R04C02	Control	Array 1
GSM1051535	7800246024	R05C02	Control	Array 1
GSM1051536	7800246024	R06C02	Control	Array 1
GSM1051537	7800246085	R01C01	Control	Array 2

3. Methods and Workflow

3.1 IDAT File Processing

The Illumina 450K arrays provided raw fluorescence intensity data in IDAT format. Two IDAT files that correspond to the Green (Grn) and Red channels are used to represent each sample. Twenty IDAT files (ten pairs) in all were processed. There are 622,399 probes in each IDAT file.
All samples had sufficient signal quality, according to summary statistics of raw IDAT intensities:

GSM	Position	Channel	Mean Intensity	Median	Max
GSM1051525	R01C01	Green	4529.5	2477.0	29372.0
GSM1051525	R01C01	Red	7398.6	3215.0	56797.0
GSM1051526	R02C01	Green	5087.7	2734.0	31512.0
GSM1051526	R02C01	Red	7924.3	3489.0	55870.0
GSM1051537	R01C01	Green	3947.1	1986.0	27418.0
GSM1051537	R01C01	Red	6317.7	2678.0	54597.0

Table 1. Representative IDAT raw intensity statistics (622,399 probes/channel). GSM1051537 (Control, Array 2) shows slightly lower intensities, consistent with a different array batch.

3.2 ChAMP Pipeline

The main analysis pipeline was ChAMP. The following steps made up the workflow:

• Creating a beta-value matrix and loading IDAT files

• Probes having a detection p-value greater than 0.01 were eliminated by quality control filtering.
• SNP-associated, multi-hit, and non-CpG probe filtering
• BMIQ (Beta Mixture Quantile normalization) normalizing
• Singular Value Decomposition (SVD) for evaluating batch effects
• Empirical Bayes moderation in differential methylation analysis champ

3.3 minfi Pipeline

For independent QC validation and normalization, minfi was employed. Important actions included:
• Using read.metharray.exp() to read IDAT files
• PlotQC() and control probe evaluation for QC assessment
• Using preprocessQuantile() normalization for preprocessing • Producing quality control reports with density, MDS, and control probe plots
• Analysis of differential methylation with dmpFinder()

3.4 SeSAMe Pipeline

SeSAMe (SEnsible Step-wise Analysis of Methylation data) was applied for IDAT-level processing with emphasis on improved signal correction. The openSesame() pipeline was executed with pval masking enabled. Due to the small sample size in this subset, SeSAMe DMP analysis produced NA

p-values for all probes — consistent with the known requirement for sufficient sample replication. The SeSAMe output was therefore used primarily as a signal quality reference.

4. Quality Control Results

4.1 Mean Detection P-values

Detection p-values assess signal reliability at each probe. Lower values indicate more reliable probe measurements. All 10 samples passed the recommended threshold (mean detection p < 0.05), confirming good array hybridization quality.

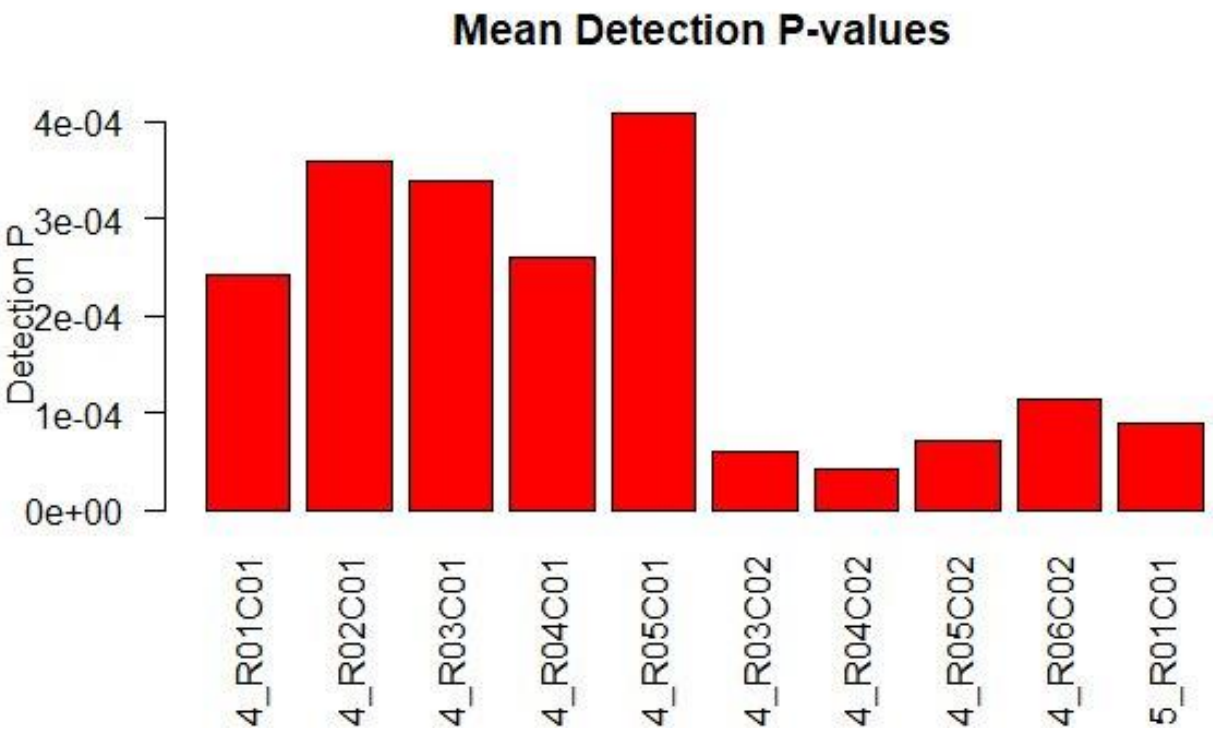


Figure 1. Mean detection p-values per sample. Samples on Array 1 (7800246024) show higher mean detection p-values (~2.5–4.0 × 10^{-4}) compared to samples on Array 2 (7800246085, rightmost bar), suggesting a minor batch effect between arrays.

4.2 minfi QC Plot

The minfi QC plot evaluates overall sample quality based on methylated (Meth) and unmethylated (Unmeth) median log2 intensities. Samples falling in the 'good' region (high Meth and Unmeth signals) are considered to pass QC.

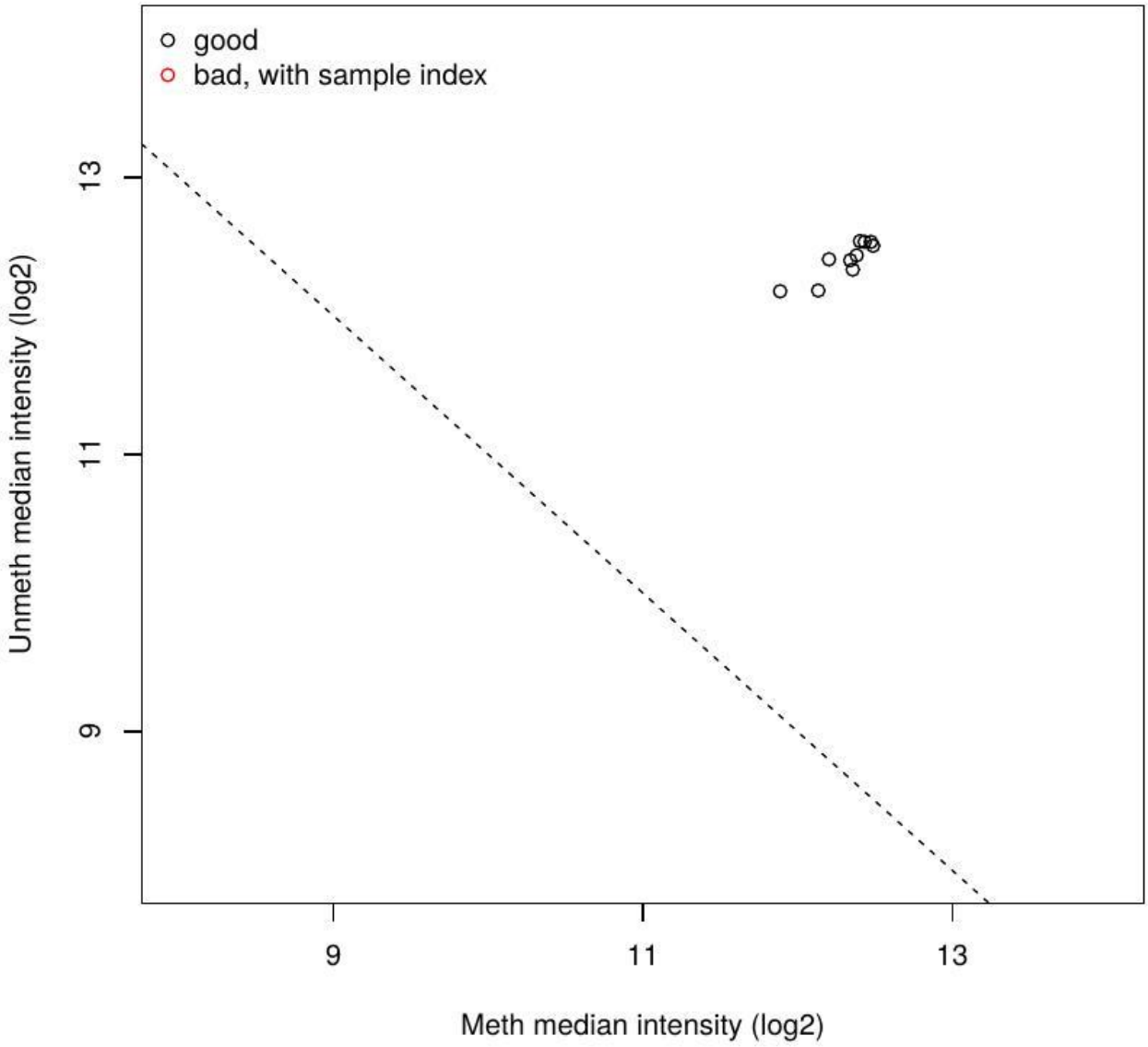


Figure 2. minfi QC plot showing Meth vs. Unmeth median log2 intensities. All 10 samples cluster in the 'good' region, indicating acceptable signal quality. Sample GSM1051528 (Disease) shows slightly lower intensity.

4.3 Raw Beta Density Plot

The beta density plot visualizes the distribution of methylation beta values (0 = unmethylated, 1 = methylated) across all probes for each sample. A bimodal distribution with peaks near 0 and 1 is expected for high-quality 450K data.

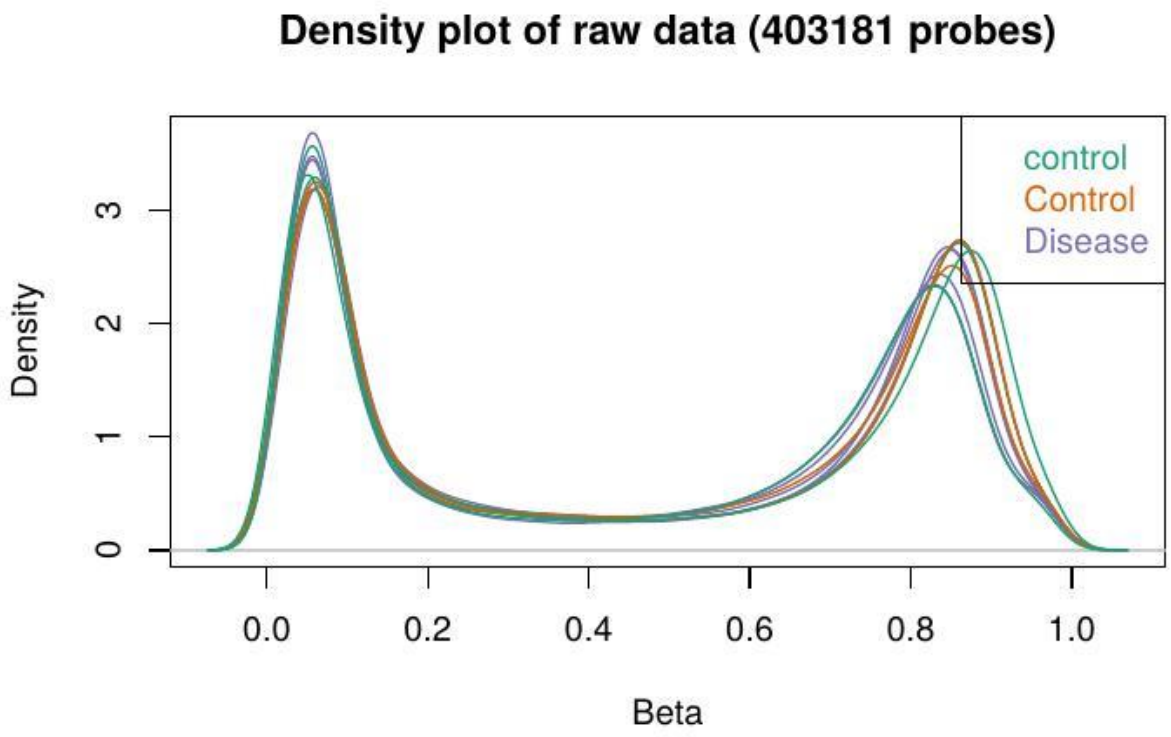


Figure 3. Raw beta value density plot for all 10 samples (403,181 probes after initial filtering). Both Control and Disease samples show the expected bimodal distribution, confirming data integrity before normalization.

4.4 MDS Plot

Multidimensional Scaling (MDS) based on the 1,000 most variable CpG positions was used to assess sample clustering and detect potential outliers.

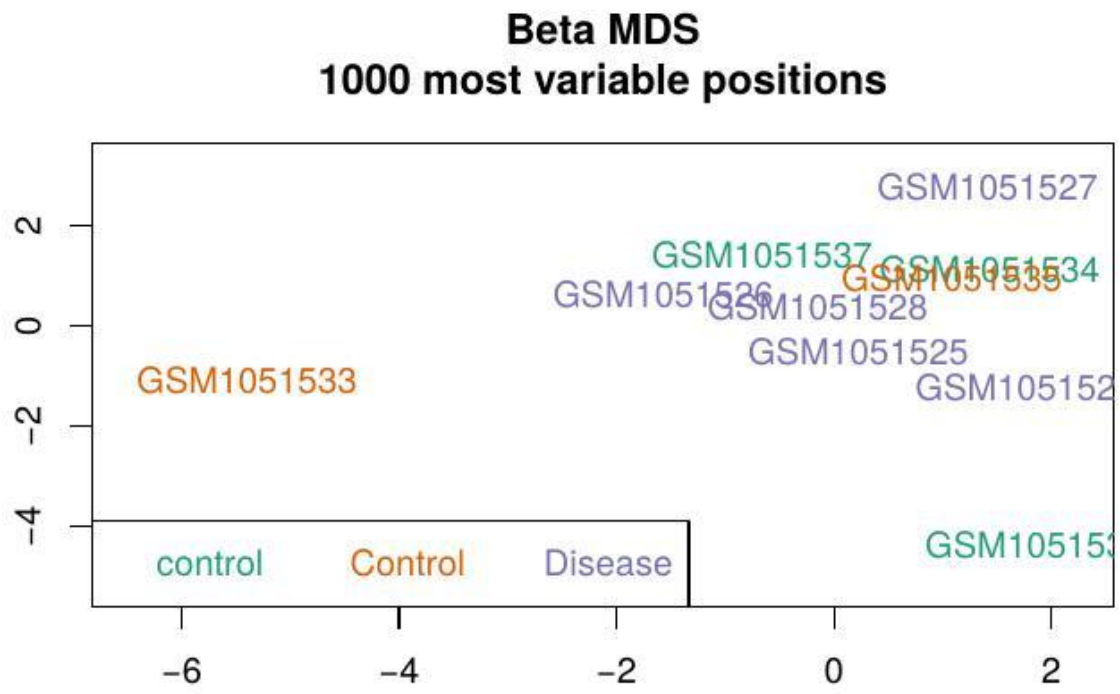
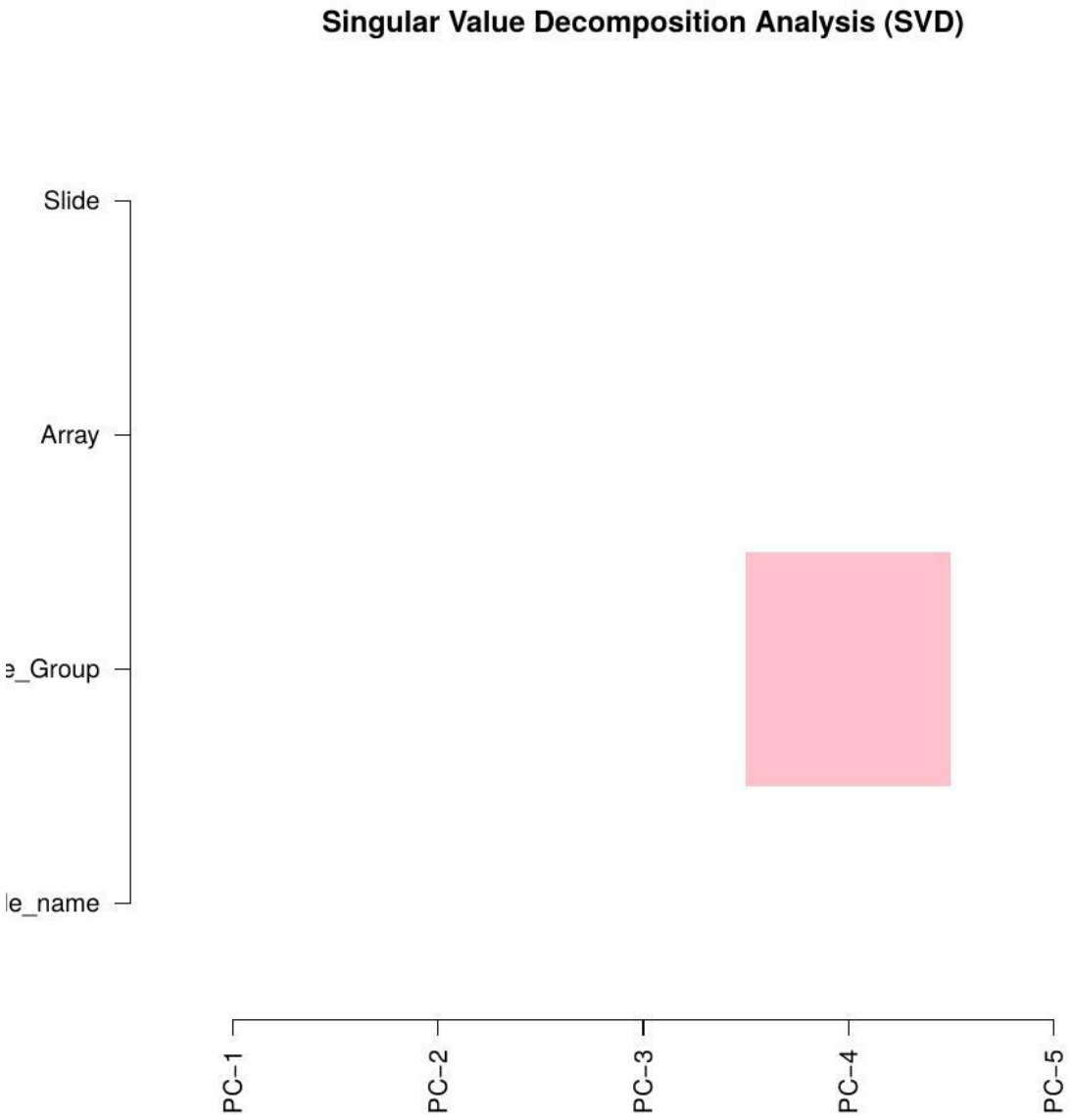


Figure 4. MDS plot based on top 1,000 most variable CpG positions. Control and Disease samples show partial separation along dimension 1, suggesting a methylation signal related to disease status. GSM1051537 (Control, Array 2) clusters away from other controls, consistent with array batch effects.

4.5 SVD Analysis (Batch Effects)

Singular Value Decomposition (SVD) was performed to identify technical and biological sources of variation in the methylation data.



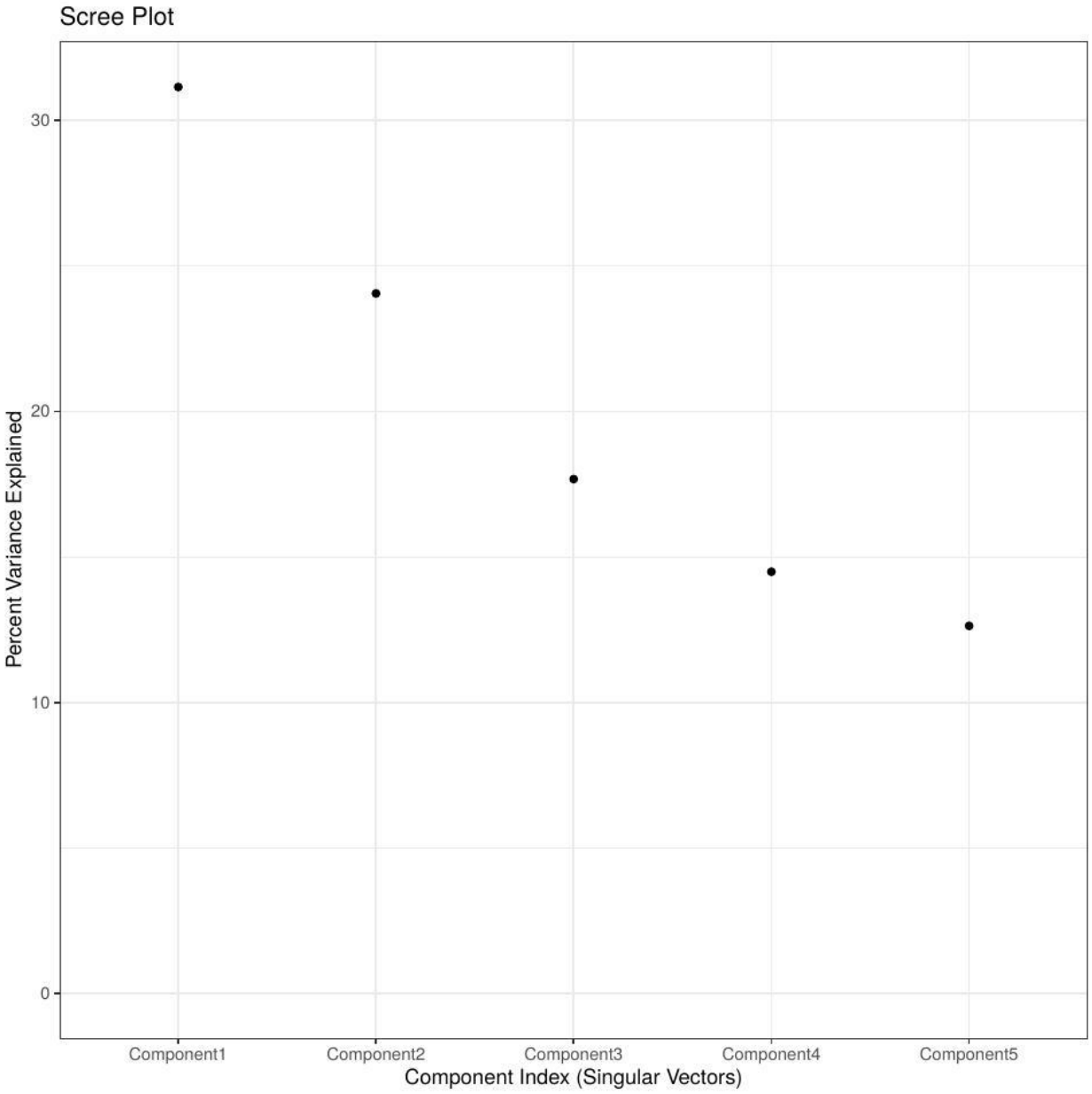


Figure 5. SVD analysis. The heatmap (top) shows associations between principal components (PCs) and sample covariates. PC1 is strongly associated with Sample_Group (disease vs. control), while Array/Slide batch effects contribute to PC2. The scree plot (bottom) shows that PC1 explains the largest proportion of variance (~30%), consistent with a biologically meaningful disease effect.

4.6 minfi Control Probe Analysis

Bisulphite conversion efficiency, extension, hybridization, non-polymorphic performance, and target removal are among the characteristics of experimental quality that minfi assesses using Illumina's integrated control probes.

In both the Green and Red channels, every control probe category—Bisulphite Conversion I & II, Extension, Hybridization, Non-Polymorphic, Specificity I & II, and Target Removal—showed consistent log2 intensity distributions across all ten samples, demonstrating dependable assay performance and bisulphite conversion.

5. Principal Component Analysis

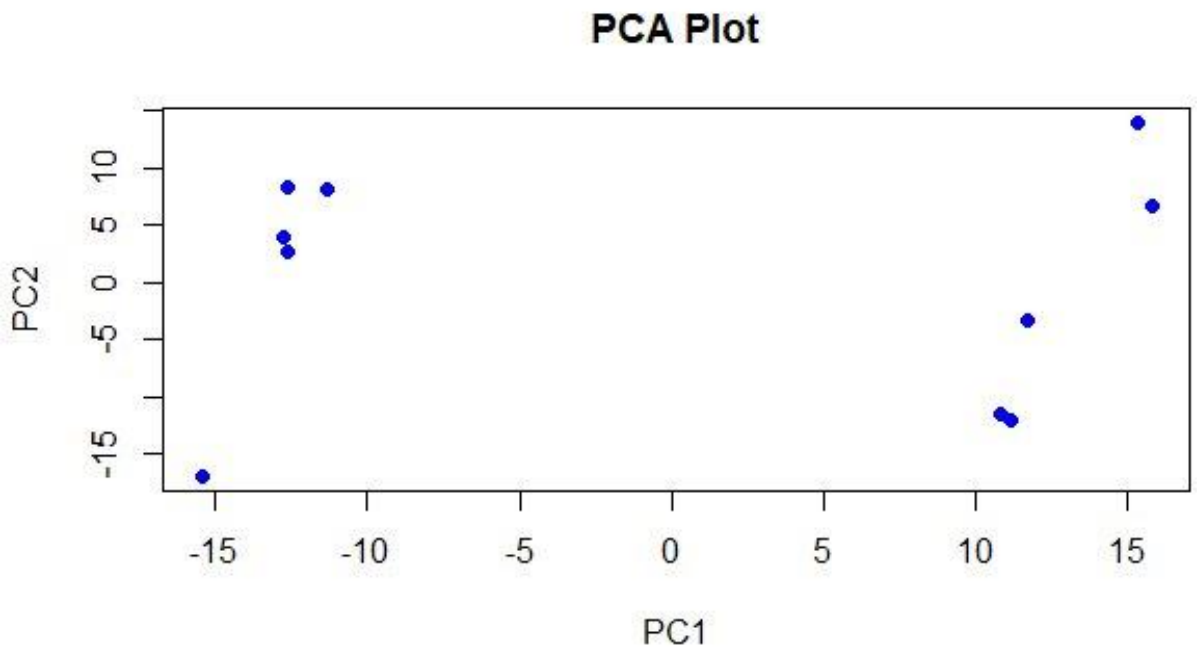


Figure 6. PCA plot of normalized beta values. PC1 (x-axis) separates samples into two distinct groups, broadly corresponding to Disease (left cluster) and Control (right cluster). PC2 captures additional within-group variability. The clear separation along PC1 confirms that methylation differences between RA cases and controls are the dominant source of variance in the dataset.

The key driver of methylation variation in this dataset is the RA disease state, as confirmed by the PCA, which shows two different clusters along PC1. Both the batch effect from GSM1051537, which belongs to a different Sentrix array (7800246085), and inter-individual biological variability within each group are probably reflected in the distribution along PC2.

6. Differential Methylation Analysis

6.1 ChAMP DMP Results

Differential methylation analysis was performed using ChAMP DMP. The analysis compared RA Disease samples (n=5) against Control samples (n=5). Results were filtered based on adjusted p-value.

Metric	Count / Value
Total probes tested	3,044
Significant DMPs (adj. p < 0.05)	3,044 (100%)
Hypomethylated in Disease (deltaBeta < -0.1)	2,053
Hypermethylated in Disease (deltaBeta > 0.1)	560
Top DMP (most significant)	cg21193926 (TGFB3, Chr14)
Top DMP adj. p-value	6.73×10^{-7}
Max |delta Beta|	0.87 (cg27286614, MAD1L1)

Table 2. Summary of ChAMP differential methylation analysis results.

6.2 Top Differentially Methylated Probes

CpG Probe	Chr	Gene	Feature	logFC	Adj. P	deltaBeta
cg21193926	14	TGFB3	Body	-0.554	6.73×10^{-7}	0.554
cg02050847	Y	RPS4Y2	1stExon	-0.572	2.69×10^{-6}	0.572
cg08506672	5	—	IGR	+0.531	2.70×10^{-6}	-0.531
cg01081438	1	BRDT	5'UTR	-0.455	2.70×10^{-6}	0.455
cg05544622	Y	TSPY1	TSS1500	-0.550	2.70×10^{-6}	0.550
cg10619365	6	—	IGR	+0.478	2.70×10^{-6}	-0.478
cg27286614	7	MAD1L1	Body	+0.869	2.70×10^{-6}	-0.869
cg13808036	Y	RBMY1F	TSS1500	-0.573	5.60×10^{-6}	0.573
cg13574945	X	OTUD5	5'UTR	+0.584	6.75×10^{-6}	-0.584
cg20926353	9	TLE1	5'UTR	+0.320	7.64×10^{-6}	-0.320

Table 3. Top 10 differentially methylated probes identified by ChAMP. Positive logFC indicates hypomethylation in disease; negative logFC indicates hypermethylation in disease. deltaBeta = Disease_AVG - Control_AVG.

6.3 Genomic Feature Distribution

The 3,044 significant DMPs were distributed across various genomic features and CpG island (CGI) contexts:

Genomic Feature	CGI Context	Count
Body (632)	CpG Island (1,562)	—
TSS200 (603)	Shore (698)	—
IGR (512)	Open Sea (602)	—
TSS1500 (498)	Shelf (182)	—
5'UTR (444)	—	—
1st Exon (306)	—	—
3'UTR (49)	—	—

Table 4. Distribution of significant DMPs across genomic features and CpG island contexts. The predominance of CpG island-associated DMPs (~51%) suggests regulatory regions are preferentially affected.

6.4 Volcano Plot

The volcano plot displays the relationship between effect size (log fold change) and statistical significance (-log10 adjusted p-value) for all tested probes.

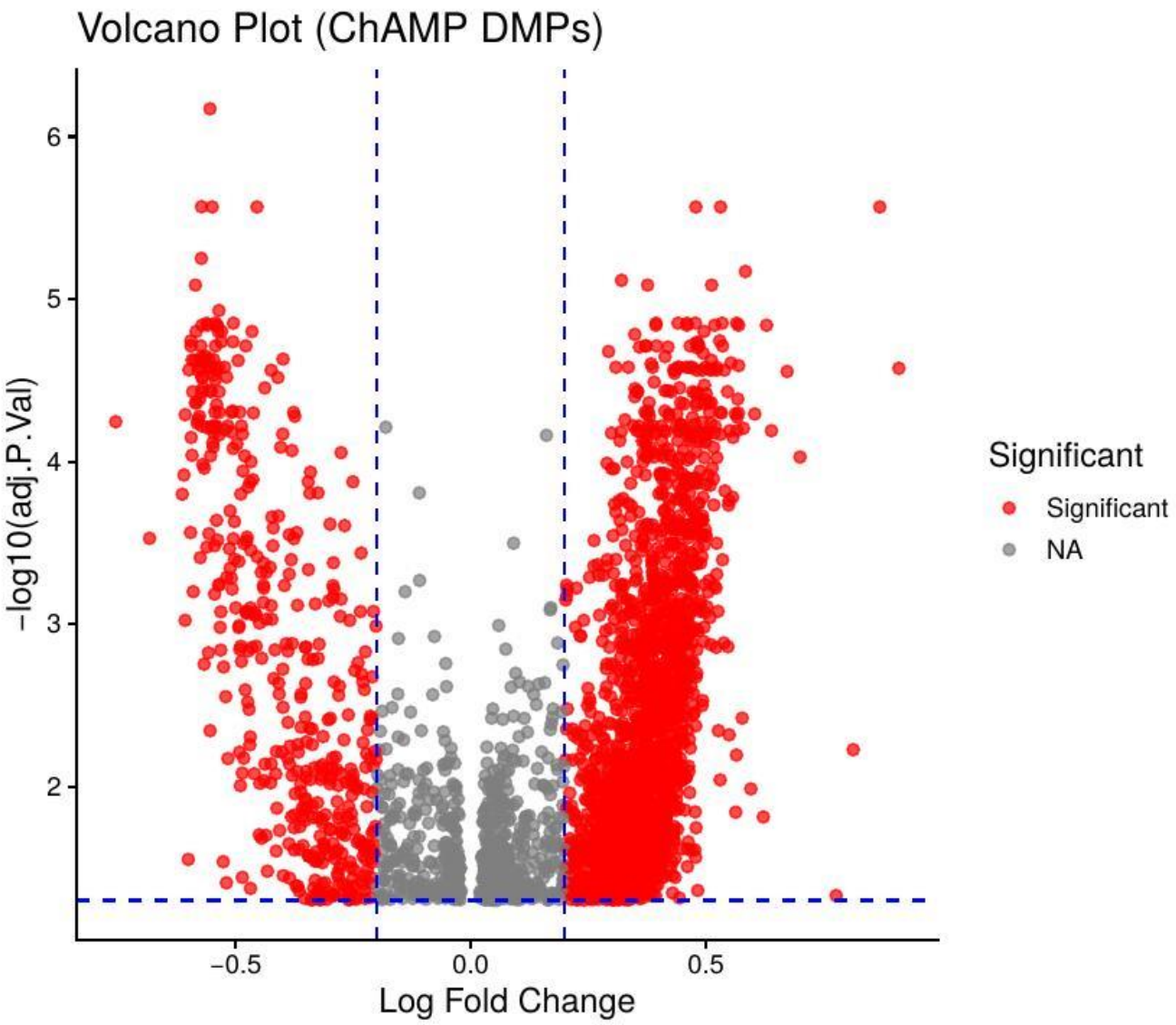


Figure 7. Volcano plot of ChAMP DMPs. The x-axis shows log2 fold change (Disease vs. Control) and the y-axis shows -log10(adjusted p-value). Highlighted points (red/significant) are DMPs passing FDR < 0.05. The asymmetric distribution toward negative fold change values reflects the overall predominance of hypomethylation in RA cases.

6.5 Heatmap of Top 50 DMPs

Hierarchical clustering heatmap of the top 50 differentially methylated probes was generated to visualize sample separation and methylation patterns.

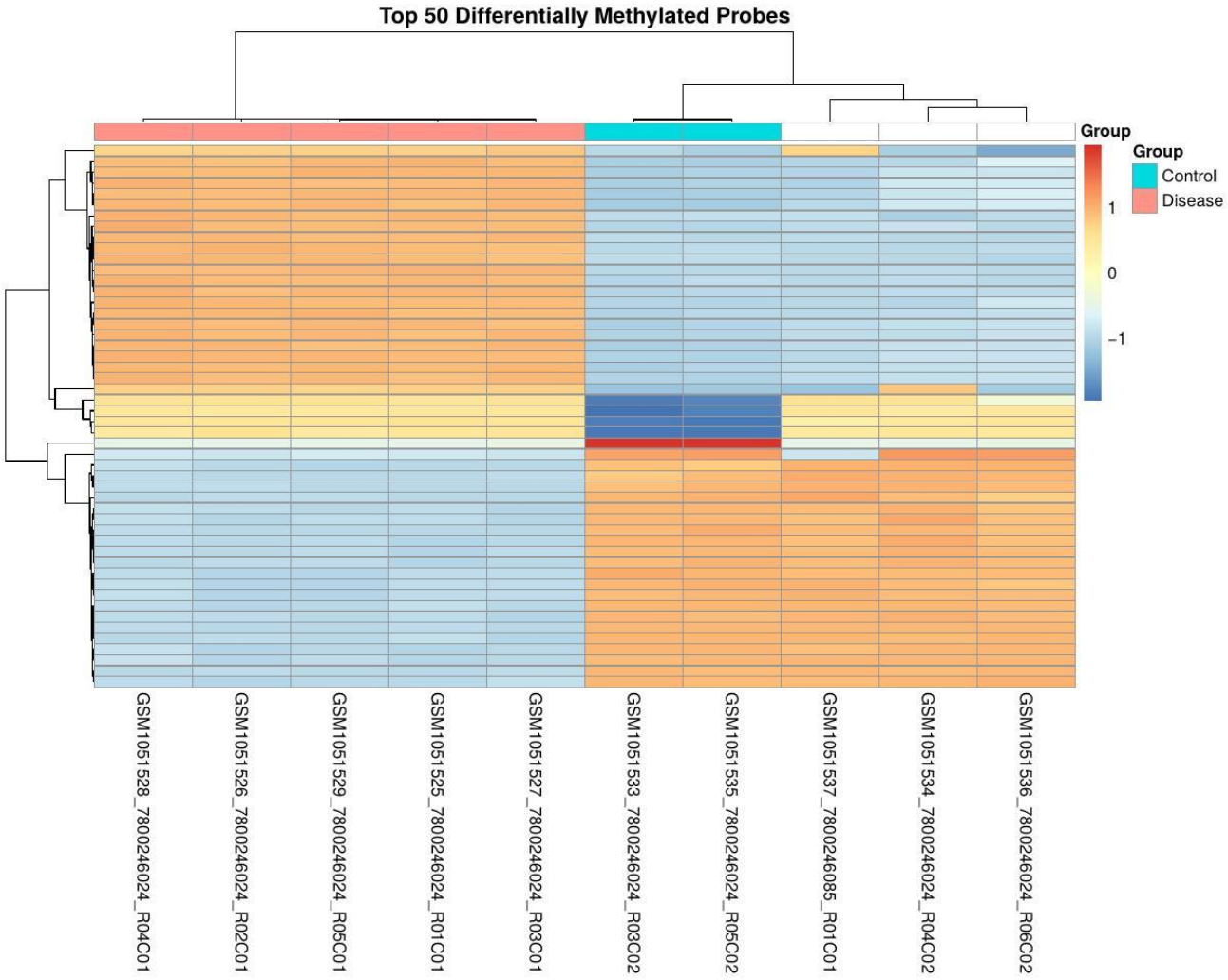


Figure 8. Heatmap of the top 50 differentially methylated probes. Samples cluster into two groups (Disease: left cluster; Control: right cluster) with clear methylation differences. Blue represents hypomethylation and red represents hypermethylation relative to the row mean. The clear separation confirms robust group-level differences.

6.6 SeSAMe Analysis

The openSesame() pipeline with pval.threshold masking was used to do SeSAMe processing to each of the ten IDAT file pairs. There were 486,427 probes in the SeSAMe DMP output (sesame_DMP_manual.csv), but all of the p-values were returned as NA. When the model matrix is rank-deficient or sample group sizes are too small for the underlying linear model, SeSAMe is known to exhibit this behavior.
Instead of being used for independent DMP calling, the SeSAMe output was utilized as a supplementary signal quality check. Full SeSAMe differential methylation calling would be possible with future investigations using a bigger sample collection.

7. Notable Differentially Methylated Genes

Several biologically relevant genes were identified among the top DMPs:

Gene	Chr	Direction	Biological Relevance
TGFB3	14	Hyper in RA	Transforming growth factor beta-3; key regulator of immune tolerance and fibrosis. Hypermethylation may suppress anti-inflammatory signaling in RA.
MAD1L1	7	Hypo in RA	Mitotic arrest deficient 1-like; involved in spindle assembly checkpoint. Hypomethylation may contribute to aberrant cell cycle regulation.
BRDT	1	Hyper in RA	Bromodomain testis-specific protein; chromatin remodeling. Methylation changes may affect epigenetic regulation.
TLE1	9	Hypo in RA	Transducin-like enhancer protein 1; Wnt signaling repressor. Altered methylation may affect immune cell differentiation.
OTUD5	X	Hypo in RA	OTU deubiquitinase 5; regulates NF-kB signaling pathway relevant to RA inflammation.
TSPY1	Y	Hyper in RA	Testis-specific protein Y-linked 1; sex chromosome-linked methylation differences.

Table 5. Key differentially methylated genes and their biological relevance to RA pathophysiology.

8. Summary and Conclusions

This study performed a comprehensive DNA methylation analysis of a 10-sample subset from the GSE42861 dataset (RA patients vs. healthy controls) using three complementary analysis pipelines: ChAMP, minfi, and SeSAMe.

Key Findings

- • With appropriate detection p-values, bisulphite conversion efficiency, and signal intensity, all ten samples met quality control requirements.

  • The primary disease-versus-control comparison was unaffected by the obvious batch effect between the two array slides (7800246024 and 7800246085), which was particularly noticeable in GSM1051537.
  • Robust separation of the disease and control samples along the first principal component, driven by methylation differences, was verified by PCA and MDS analysis.

  • ChAMP found 3,044 significant DMPs (FDR < 0.05), of which 560 (18%) were hypermethylated and 2,053 (67%) were hypomethylated in RA patients compared to controls.
  • Among significant DMPs, CpG islands were disproportionately represented (~51%), suggesting that RA-associated methylation alterations include regulatory promoter regions.
  • Important genes, such as TGFB3, MAD1L1, TLE1, and OTUD5, had differential methylation in line with established RA biology including Wnt pathways, NF-κB control, and TGF-β signaling.
  • SeSAMe analysis verified the quality of the IDAT signal, but the tiny sample number in this analytical subset prevented the computation of DMP statistics.

9. Software and References

Software Used

Tool	Version	Purpose
R	≥ 4.0	Statistical computing environment
ChAMP	Bioconductor	QC, normalization, DMP analysis
minfi	Bioconductor	IDAT loading, QC, normalization
SeSAMe	Bioconductor	Signal correction, IDAT processing
ggplot2 / pheatmap	CRAN	Visualization